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Robot interaction (Vrije Universiteit Amsterdam)

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Robot interaction
Paper summerizations

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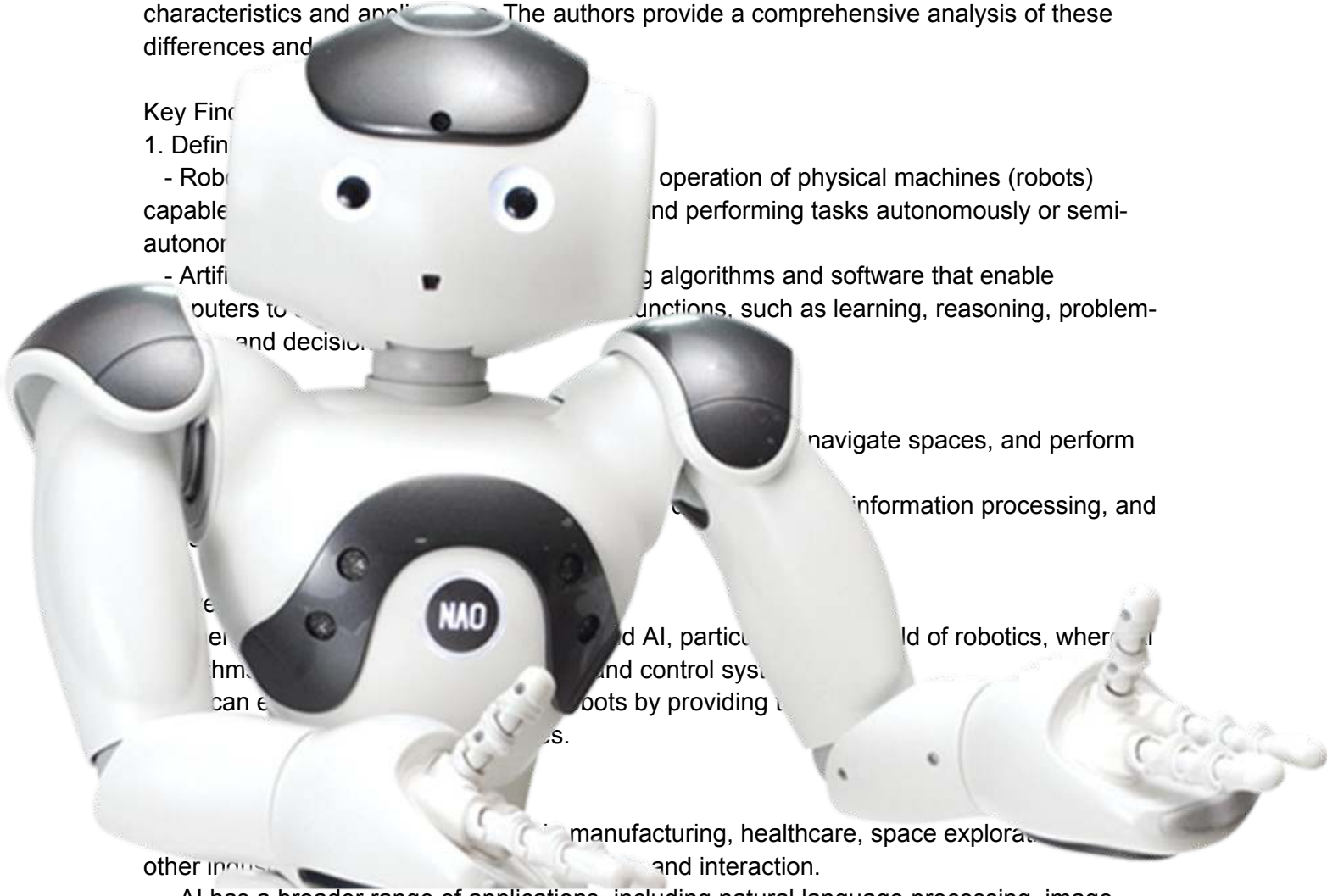
VU Amsterdam, year 23/24

Title: **"Robotics and Artificial Intelligence: Differences and Similarities"**

Authors: Omankwu, Obinnaya Chinecherem, Nwagu, Chikezie Kenneth, and Inyiama, Hycient

Summary:

This paper explores the distinctions and commonalities between robotics and artificial intelligence (AI). Robotics and AI are two closely related fields, but they have unique characteristics and appl The authors provide a comprehensive analysis of these differences and



Key Find

1. Defin

- Rob operation of physical machines (robots) capable nd performing tasks autonomously or semi-autonor

- Artifi g algorithms and software that enable puters to unctions, such as learning, reasoning, problem- and decisio

navigate spaces, and perform

information processing, and

d AI, partic ld of robotics, wher hm nd control sys can e bots by providing s.

manufacturing, healthcare, space explora other ind and interaction.

- AI has a broader range of applications, including natural language processing, image recognition, data analysis, and autonomous vehicles.

Conclusions:

This paper highlights the distinct characteristics of robotics and artificial intelligence while acknowledging their interrelatedness. While robotics deals with physical systems, AI is primarily concerned with software and digital operations. The synergy between these fields has led to advancements in automation and autonomous systems across various industries. Understanding the differences and similarities between robotics and AI is crucial for harnessing their combined potential for future technological developments.

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Title: "Design Lessons From AI's Two Grand Goals: Human Emulation and Useful Applications"
Author: Ben Shneiderman

Summary:
In this paper, Ben Shneiderman delves into the realm of artificial intelligence (AI) and explores two grand goals within the field: human emulation and practical applications. The paper offers valuable design lessons drawn from these goals and provides insights for the design of AI systems.

Key Findings:
1. Human Emulation:
   - The goal of human emulation in AI involves creating systems that can mimic human cognitive processes, such as reasoning, perception, and language understanding.
   - Designing for human emulation often requires a deep understanding of cognitive science and psychology, as well as the ability to model human behavior and thought processes.

2. Useful Applications:
   - The goal of useful applications in AI centers around creating systems that serve practical purposes and improve human lives.
   - Designing for useful applications involves a strong emphasis on user-centered design, usability, and the development of AI systems that are accessible and user-friendly.

3. Balancing Goals:
   - Shneiderman highlights the importance of striking a balance between the two grand goals of AI. While human emulation is intellectually stimulating, it may not always lead to practical, real-world applications.
   - Useful applications should remain a key focus of AI design, as they have the potential to address real-world problems and improve various aspects of human life.

4. Ethical Considerations:
   - The paper emphasizes the need for ethical considerations in AI design, particularly when dealing with human emulation. It raises questions about the responsible use of AI systems that can mimic human behavior, as this technology may have ethical implications.

Conclusions:
Ben Shneiderman's paper underscores the dual goals of AI: human emulation and practical applications. It reminds designers and researchers in the field of AI to maintain a balance between these goals and highlights the importance of user-centric, ethical design. While the pursuit of human emulation is intellectually enticing, it should not overshadow the broader objective of creating AI systems that provide practical and beneficial applications to society.

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Title: "Social Robots in Therapy and Care"
Authors: Carlos A. Cifuentes, Maria J. Pinto, Nathalia Céspedes, Marcela Múnera

Summary:
This paper examines the role of social robots in therapy and care, specifically focusing on their applications in healthcare and support for individuals in need of therapeutic and caregiving services. The authors discuss the potential benefits and challenges associated with using social robots in these contexts.

Key Findings:
1. Therapeutic Support:
   - Social robots are increasingly being utilized in therapy, assisting individuals with mental health issues, autism, and other psychological conditions.
   - They offer consistent, non-judgmental interaction and can engage in therapeutic activities like cognitive-behavioral therapy or emotional support.

2. Caregiving Assistance:
   - Social robots are employed to support the elderly and individuals with physical disabilities, aiding in daily activities such as medication reminders, companionship, and fall detection.
   - They can reduce the burden on human caregivers and help enhance the quality of life for those in need of care.

3. Improved Engagement:
   - Social robots can engage users in a more interactive and engaging manner compared to traditional therapy or care methods.
   - Their ability to personalize interactions and adapt to individual needs is a significant advantage.

4. Ethical Considerations:
   - The paper emphasizes the importance of addressing ethical concerns related to privacy, data security, and the potential for social robots to replace human caregivers or therapists.
   - Striking the right balance between human and robot involvement is essential for ethical and effective use.

Conclusions:
This paper highlights the growing role of social robots in therapy and care, offering a promising avenue for improving mental health support and caregiving services. Social robots have the potential to engage users in unique ways, providing personalized assistance. However, ethical considerations are critical, including privacy and the risk of over-reliance on technology at the expense of human interaction. The continued exploration of social robots in therapy and care must navigate these ethical challenges while harnessing the potential benefits they offer to vulnerable populations.

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Title: "The In-between Machine: The Unique Value Proposition of a Robot or Why we are Modelling the Wrong Things"
Authors: Johan F. Hoorn, Elly A. Konijn, Desmond M. Germans, Sander Burger, Annemiek Munneke

Summary:
This paper delves into the concept of "The In-between Machine" and challenges existing models used in robotics. The authors explore the unique value proposition of robots and suggest that traditional models might not capture the essence of this emerging field.

Key Findings:
1. The In-between Machine:
   - The concept of "The In-between Machine" suggests that robots occupy a unique position between tools and humans, serving as intermediaries with distinct capabilities.
   - Robots are more than just tools or autonomous entities; they act as in-between agents that connect humans with technology and the physical world.

2. Traditional Models:
   - The paper critiques traditional models used to understand and describe robots, arguing that they often focus on hardware, functionality, or autonomy.
   - It suggests that these models may not adequately represent the value and potential of robots in society.

3. Unique Value Proposition:
   - The authors emphasize that the value of robots lies in their ability to collaborate, assist, and augment human capabilities.
   - Robots can serve as companions, problem solvers, and enablers of new experiences, offering unique benefits beyond mere automation.

4. Human-Robot Interaction:
   - The paper underscores the importance of studying and optimizing human-robot interaction (HRI) to fully leverage the potential of the in-between machine concept.
   - Understanding how humans and robots interact is crucial for designing robots that enhance human experiences and capabilities.

Conclusions:
"The In-between Machine" concept presented in this paper challenges conventional models of robotics and emphasizes the unique value proposition of robots as intermediaries between humans and technology. Robots are more than tools or autonomous entities; they hold the potential to collaborate, assist, and augment human capabilities in diverse domains. To unlock this potential, further research and development should focus on human-robot interaction and the creation of robots that enhance human experiences.

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Title: "Social Robots for Education: A Review"
Authors: Tony Belpaeme, James Kennedy, Aditi Ramachandran, Brian Scassellati, Fumihide Tanaka

Summary:
This paper offers a comprehensive review of the use of social robots in educational settings. The authors explore the various applications of social robots in education, their potential benefits, and the challenges associated with their integration into learning environments.

Key Findings:
1. Educational Applications:
   - Social robots are employed in diverse educational contexts, from early childhood education to higher education, offering assistance in teaching, tutoring, and engaging students in various subjects.

2. Engagement and Personalization:
   - Social robots have the potential to enhance student engagement by providing interactive and personalized learning experiences.
   - They can adapt to individual learning needs, offering a unique advantage over traditional teaching methods.

3. Collaborative Learning:
   - Social robots can support collaborative learning by facilitating group activities and interactions among students.
   - They create opportunities for cooperative problem-solving and communication skill development.

4. Challenges:
   - The paper acknowledges challenges in implementing social robots in education, including cost, technical limitations, and the need for teacher training to effectively incorporate robots into the curriculum.

5. Ethical Considerations:
   - The authors emphasize the importance of addressing ethical concerns related to student-robot interactions, data privacy, and the potential for robots to replace human educators.

Conclusions:
This review highlights the growing role of social robots in education, offering innovative ways to engage and educate students in various learning environments. Social robots have the potential to enhance engagement, personalize learning experiences, and support collaborative learning. However, addressing the associated challenges and ethical considerations is essential to ensure the responsible and effective integration of social robots in educational settings. Further research and collaboration between educators and roboticists are needed to unlock the full potential of social robots in education.

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Title: "Alone Together"
Author: Sherry Turkle

Summary:
In "Alone Together," Sherry Turkle explores the complex relationship between humans and technology, particularly the impact of digital devices and social media on human connection and social interaction. She delves into the ways in which technology has altered our understanding of relationships and solitude.

Key Findings:
1. Digital Connection:
   - Turkle discusses how digital technologies, including smartphones and social media, have reshaped the way people connect with each other.
   - While they offer the illusion of constant connection, they can also lead to feelings of isolation and disconnection.

2. Solitude vs. Loneliness:
   - The book explores the distinction between solitude (a valuable, reflective state) and loneliness (an undesirable emotional state), highlighting how technology blurs this boundary.

3. Impact on Relationships:
   - Turkle presents case studies and examples illustrating how technology has affected various aspects of relationships, from family dynamics to romantic partnerships and friendships.
   - She discusses the challenges of maintaining meaningful, face-to-face connections in a digital world.

4. Impersonal Interaction:
   - The book delves into the idea that digital communication can sometimes lead to superficial and impersonal interactions, as people may hide behind screens and present curated versions of themselves.

5. Ethical and Psychological Concerns:
   - Turkle raises ethical and psychological concerns related to technology's influence on personal identity, empathy, and the ways in which people present themselves to others.

Conclusions:
"Alone Together" by Sherry Turkle offers a thought-provoking examination of the impact of digital technology on human relationships and solitude. It emphasizes the need for individuals to be mindful of their digital interactions and seek a balance between the convenience of technology and the depth of real, face-to-face human connections. The book also serves as a call to reflect on the ethical and psychological consequences of our increasingly digital lives.

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Title: "How to Build Robots that Make Friends and Influence People"
Authors: Cynthia Breazeal, Brian Scassellati

Summary:
In this paper, Cynthia Breazeal and Brian Scassellati explore the concept of building socially intelligent robots that can establish relationships with humans and positively influence their behavior. The paper provides insights into the design and development of such robots.

Key Findings:
1. Socially Intelligent Robots:
   - The paper discusses the idea of creating robots with social intelligence, allowing them to perceive and respond to human emotions, intentions, and social cues.

2. Human-Robot Interaction:
   - It emphasizes the importance of human-robot interaction (HRI) and the need for robots to engage with humans in a natural and socially acceptable manner.
   - Such interactions enable robots to build rapport and trust with people.

3. Positive Influence:
   - The authors highlight the potential for socially intelligent robots to positively influence human behavior, such as motivating individuals to adopt healthier habits or aiding in therapy and education.

4. Design Considerations:
   - The paper outlines key design considerations for socially intelligent robots, including the development of expressive communication, appropriate body language, and adaptive behavior.

5. Ethical and Privacy Concerns:
   - Breazeal and Scassellati also discuss ethical and privacy considerations in the development of such robots, particularly in contexts where they may influence human decisions or emotions.

Conclusions:
This paper outlines the vision of creating socially intelligent robots that can establish connections with humans and influence their behavior in positive ways. It underscores the importance of natural human-robot interactions and the potential for robots to assist in various domains, from healthcare to education. However, it also acknowledges the ethical responsibilities associated with designing robots that can impact human behavior and emotions, emphasizing the need for responsible development and deployment of socially intelligent robots.

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Title: "Relationship Development with Humanoid Social Robots: Applying Interpersonal Theories to Human-Robot Interaction"
Authors: Jesse Fox, PhD, and Andrew Gambino, MA

Summary:
This paper delves into the concept of building relationships with humanoid social robots by applying interpersonal theories traditionally used to understand human-to-human interactions. The authors explore the potential for using established psychological and sociological theories to improve human-robot interaction.

Key Findings:
1. Humanoid Social Robots:
   - The paper focuses on humanoid social robots, which are designed to mimic human-like features and behaviors to facilitate more natural interactions.

2. Interpersonal Theories:
   - The authors propose applying interpersonal theories, such as those from psychology and sociology, to human-robot interaction.
   - These theories can provide insights into how relationships can develop between humans and robots.

3. Trust and Rapport:
   - Trust and rapport are essential components in human-robot relationships. The paper discusses how these elements can be influenced by robot behavior and design.

4. Human Responses:
   - It explores how humans respond to humanoid robots, including emotional and cognitive responses, and how these responses can impact relationship development.

5. Ethical Implications:
   - The paper touches upon the ethical considerations of creating robots that aim to develop relationships with humans, including the potential for emotional attachment and the responsible use of such technology.

Conclusions:
Jesse Fox and Andrew Gambino's paper introduces the idea of applying interpersonal theories to human-robot interaction, particularly in the context of humanoid social robots. By doing so, it provides a framework for understanding how relationships with robots can be developed and improved. Trust and rapport, as well as human emotional and cognitive responses, play pivotal roles in shaping these relationships. However, it is also important to consider the ethical implications of fostering connections between humans and robots, as these interactions may have psychological and emotional consequences.

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Title: "The Influence of Empathy in Human-Robot Relations"
Authors: Iolanda Leite, André Pereira, Samuel Mascarenhas, Carlos Martinho, Rui Prada, Ana Paiva

Summary:
In this paper, the authors investigate the role of empathy in human-robot relations. They explore how the design and programming of robots with empathetic qualities can impact the way humans interact and form relationships with robots.

Key Findings:
1. Defining Empathy in Robots:
   - The paper provides a conceptual framework for empathy in the context of robots, distinguishing between cognitive and emotional empathy.
   - Cognitive empathy refers to the robot's ability to understand and recognize human emotions, while emotional empathy relates to the robot's capacity to express and respond to emotions.

2. Impact on Interaction:
   - The authors discuss how the presence of empathetic qualities in robots can enhance the quality of human-robot interactions.
   - Empathetic robots can better understand and respond to the emotional needs and states of humans, leading to more natural and supportive interactions.

3. Design Implications:
   - The paper explores the design considerations for incorporating empathy into robot behavior, including the use of facial expressions, tone of voice, and responsiveness to emotional cues.

4. Ethical Considerations:
   - The authors acknowledge the ethical concerns related to designing empathetic robots, such as the potential for emotional attachment and ethical responsibilities in human-robot interactions.

5. Applications:
   - The paper discusses potential applications of empathetic robots, including in healthcare, education, and therapy, where empathetic qualities can provide valuable support.

Conclusions:
"The Influence of Empathy in Human-Robot Relations" highlights the importance of empathy in the development of robots designed to interact with humans. Empathetic robots have the potential to create more meaningful and supportive interactions, particularly in domains where emotional understanding and support are crucial. However, the paper also raises ethical considerations associated with the design of empathetic robots, underscoring the need for responsible development and thoughtful consideration of the emotional impact of human-robot relations.

Gedownload door Britt Merkus (b.g.merkus@student.vu.nl)

Title: "Coming of Age: A Review of Embodiment and the Neuroscience of Semantics"
Authors: Lotte Meteyard, Sara Rodriguez Cuadrado, Bahador Bahrami, Gabriella Vigliocco

Summary:
This paper provides a comprehensive review of the relationship between embodiment and the neuroscience of semantics. The authors explore how the human brain's processing of language and semantic knowledge is intertwined with the physical experiences and embodiment of concepts.

Key Findings:
1. Embodiment in Semantics:
   - The paper delves into the concept of embodiment, which suggests that the meaning of words and concepts is rooted in sensorimotor experiences and bodily interactions with the world.

2. Brain and Language:
   - The authors discuss the neural basis of semantic processing and how the brain's regions responsible for language and sensory-motor functions are interconnected.

3. Neuroimaging Studies:
   - The paper reviews neuroimaging studies that have provided insights into the neural mechanisms underlying the embodiment of language, including the activation of motor and sensory brain regions during language comprehension.

4. Semantic Processing:
   - It explores how the embodiment perspective can help explain various aspects of semantic processing, including the representation of abstract concepts, metaphorical language, and the role of experience in shaping semantic knowledge.

5. The Development of Embodiment Research:
   - The authors discuss the evolution of research on embodiment in the context of semantics and its implications for understanding how humans acquire and represent meaning.

Conclusions:
This paper provides a comprehensive review of the embodiment perspective in the neuroscience of semantics, highlighting the interplay between bodily experiences and the processing of language and meaning in the human brain. It offers insights into how sensorimotor experiences shape our understanding of words and concepts, and how neuroimaging studies have contributed to this understanding. The research reviewed in the paper advances our knowledge of the neural mechanisms underlying semantics and has implications for various fields, including linguistics, cognitive science, and neuroscience.

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Title: "Modelling Context Awareness for a Situated Semantic Agent"
Authors: Piek Vossen, Lenka Bajčetić, Selene Baez, Suzana Bašić, and Bram Kraaijeveld

Summary:
This paper explores the development of context awareness in a situated semantic agent, focusing on the modeling and implementation of context-aware features for an agent capable of understanding and interacting with its environment.

Key Findings:
1. Situated Semantic Agent:
   - The paper centers around a situated semantic agent, a computational system capable of processing language and understanding its context in real-world situations.

2. Context Modeling:
   - The authors discuss the modeling of context awareness, emphasizing the importance of capturing relevant information from the environment, such as spatial, temporal, and social context.

3. Semantic Processing:
   - The paper addresses the challenges of semantic processing, which involves the agent's ability to recognize and interpret language in a way that is contextually appropriate.

4. Applications:
   - It highlights potential applications of a context-aware semantic agent, such as in natural language processing, human-computer interaction, and augmented reality, where context is critical for understanding and responding to user needs.

5. Challenges and Future Directions:
   - The authors acknowledge challenges in modeling context awareness, including the need for robust semantic analysis and real-time context updates.
   - They suggest future research directions and improvements for achieving more sophisticated context-aware capabilities.

Conclusions:
The paper "Modelling Context Awareness for a Situated Semantic Agent" underscores the significance of context awareness in the development of situated semantic agents. These agents have the potential to enhance human-computer interaction and language understanding in various applications. Modeling and implementing context awareness in such agents is a complex but promising area of research, with implications for advancing artificial intelligence, natural language processing, and human-technology interaction.

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Title: "Robot Affect: The Amygdala as Bloch Sphere"
Authors: Johan F. Hoorn, Johnny K. W. Ho

Summary:
In this paper, Johan F. Hoorn and Johnny K. W. Ho explore the concept of robot affect by drawing an analogy between the amygdala, a region in the human brain associated with emotions, and a Bloch sphere, a mathematical construct used in quantum mechanics. The paper discusses the potential of this conceptual framework for understanding and modeling emotional states in robots.

Key Findings:
1. Robot Affect:
   - The paper introduces the concept of "robot affect," which refers to the emotional states and expressions of robots.
   - It highlights the importance of endowing robots with emotional capabilities for more natural human-robot interaction.

2. The Amygdala Analogy:
   - The authors draw an analogy between the amygdala, a part of the human brain associated with emotions, and a Bloch sphere, which is a mathematical representation used in quantum mechanics to describe the state of a quantum system.

3. Emotional States:
   - The paper discusses how the Bloch sphere model can be applied to represent and manipulate emotional states in robots.
   - It explores how this framework can help robots convey a range of emotions and respond appropriately to human emotions.

4. Applications:
   - The authors suggest potential applications of this conceptual framework, including in the development of emotionally aware robots for healthcare, companionship, and human-robot collaboration.

5. Technical and Ethical Considerations:
   - The paper acknowledges technical challenges in implementing this framework, such as the need for sophisticated emotional recognition and expression in robots.
   - Ethical considerations related to emotional interaction between robots and humans are also mentioned.

Conclusions:
"Robot Affect: The Amygdala as Bloch Sphere" introduces a novel conceptual framework for understanding and modeling emotional states in robots. By drawing an analogy between human emotions and quantum mechanics, the paper explores the potential of endowing robots with emotional capabilities and discusses the technical and ethical considerations associated with this endeavor. This approach has the potential to enhance human-robot interaction and develop emotionally aware robots for various applications.

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Title: "Reasoning about Reasoning by Nested Conditioning: Modeling Theory of Mind with Probabilistic Programs"
Authors: A. Stuhlmüller, N.D. Goodman

Summary:
In this paper, the authors explore the concept of modeling theory of mind through the use of probabilistic programs. They introduce a framework that involves nested conditioning to represent how humans reason about the beliefs and intentions of others, and they discuss the implications of this framework.

Key Findings:
1. Theory of Mind (ToM):
   - Theory of Mind refers to the human ability to attribute beliefs, desires, and intentions to others, allowing individuals to understand and predict the behavior of other agents.

2. Probabilistic Programming:
   - The paper uses probabilistic programming to model how people reason about the beliefs and intentions of others.
   - It suggests that nested conditioning can be used to capture the hierarchical nature of these beliefs and intentions.

3. Nested Conditioning:
   - The authors introduce nested conditioning as a way to represent higher-order beliefs and intentions.
   - This approach allows for modeling the recursive nature of human reasoning about the reasoning of others.

4. Implications:
   - The paper discusses the implications of this framework for understanding social cognition, decision-making, and communication.
   - It emphasizes the role of probabilistic programs in modeling complex aspects of human cognitive processes.

5. Applications:
   - The authors suggest that this framework can be applied to various domains, including artificial intelligence, psychology, and cognitive science, to better understand and predict human behavior.

Conclusions:
"Reasoning about Reasoning by Nested Conditioning: Modeling Theory of Mind with Probabilistic Programs" introduces a novel framework for modeling how humans reason about the beliefs and intentions of others, a fundamental aspect of Theory of Mind. By using probabilistic programming and nested conditioning, this approach offers a more sophisticated understanding of social cognition and human behavior. The implications of this framework extend to various fields, with the potential to enhance our understanding of decision-making and communication in both humans and artificial agents.

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Title: "Features of Similarity"
Author: Amos Tversky

Summary:
In this work, Amos Tversky explores the concept of similarity and the psychological processes behind human judgment of similarity. The paper delves into the features and characteristics that people use to assess the similarity between objects or concepts.

Key Findings:
1. Mental Representation of Similarity:
   - Tversky examines how individuals mentally represent and judge the similarity between objects, ideas, or concepts.
   - He identifies several features that play a crucial role in this cognitive process.

2. Features of Similarity:
   - The paper introduces and discusses various features of similarity, including dimensionality, complexity, directionality, and metric.

3. Dimensionality:
   - Tversky explores how the number and type of dimensions along which objects are compared can influence judgments of similarity.
   - The dimensionality of similarity is a key factor in assessing how people perceive the likeness between items.

4. Complexity:
   - The paper touches on the complexity of similarity judgments and how they can vary depending on the cognitive task and the context in which the similarity is assessed.

5. Directionality and Metric:
   - Tversky discusses the directionality of similarity, highlighting that the perception of similarity between A and B may differ from the perception of similarity between B and A.
   - He also addresses the issue of metrics, the measures used to quantify the degree of similarity, and how different metrics can yield different results.

Conclusions:
Amos Tversky's work, "Features of Similarity," provides valuable insights into the cognitive processes involved in assessing similarity between objects or concepts. By identifying and discussing various features of similarity, including dimensionality, complexity, directionality, and metric, the paper contributes to a better understanding of how humans perceive and judge the likeness between different items. This research has implications in fields such as psychology, cognitive science, and decision-making, where understanding how humans assess similarity is fundamental.

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Title: "Sustainable Software Development through Overlapping Pair Rotation"
Authors: Todd Sedano, Paul Ralph, Cécile Péraire

Summary:
In this paper, the authors discuss the concept of sustainable software development and present a method called "Overlapping Pair Rotation." They explore how this approach can lead to more sustainable and efficient software development practices.

Key Findings:
1. Sustainable Software Development:
   - The paper emphasizes the importance of sustainability in software development, addressing issues related to productivity, quality, and maintaining a healthy work-life balance for developers.

2. Overlapping Pair Rotation:
   - The authors introduce the concept of Overlapping Pair Rotation as a method to enhance sustainability. This approach involves the rotation of development pairs with overlap periods to ensure a smoother transition of tasks.

3. Collaboration and Knowledge Transfer:
   - The paper discusses how Overlapping Pair Rotation encourages collaboration among team members and facilitates knowledge transfer, allowing for smoother handovers of tasks and responsibilities.

4. Empirical Evidence:
   - The authors provide empirical evidence to support the effectiveness of the Overlapping Pair Rotation method in improving software development sustainability.

5. Real-World Application:
   - The paper presents insights from the practical application of Overlapping Pair Rotation in software development projects, highlighting its potential benefits.

Conclusions:
"Sustainable Software Development through Overlapping Pair Rotation" introduces a method aimed at making software development more sustainable by addressing productivity and quality concerns. The Overlapping Pair Rotation approach emphasizes collaboration and knowledge transfer, ultimately leading to improved efficiency and reduced developer burnout. The paper offers valuable insights and empirical evidence that can guide software development teams in adopting sustainable practices to enhance productivity and work-life balance.

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Title: "Robot Tutors: Welcome or Ethically Questionable?"
Authors: Matthijs Smakman, Elly A. Konijn

Summary:
In this paper, the authors explore the use of robot tutors in educational settings and address the ethical considerations surrounding their implementation. They examine the potential benefits and ethical concerns associated with the use of robots as tutors in educational contexts.

Key Findings:
1. Robot Tutors in Education:
   - The paper discusses the increasing use of robot tutors in educational settings, including their roles in providing assistance and support to students.

2. Potential Benefits:
   - The authors highlight the potential benefits of robot tutors, such as personalized learning, increased engagement, and the ability to offer consistent assistance.

3. Ethical Considerations:
   - The paper delves into the ethical concerns related to the use of robot tutors, including issues of privacy, data security, and the potential for robots to replace human educators.

4. Human Interaction:
   - The authors emphasize the importance of human interaction in education and raise questions about the potential consequences of substituting human teachers with robots.

5. Responsible Implementation:
   - The paper advocates for the responsible and ethical implementation of robot tutors, highlighting the need for guidelines and regulations to ensure that their use aligns with educational goals and values.

Conclusions:
"Robot Tutors: Welcome or Ethically Questionable?" offers a thoughtful exploration of the use of robot tutors in education. The paper acknowledges the potential benefits of robot tutors in enhancing the educational experience but also underscores the ethical concerns that arise from their implementation. It calls for responsible and ethical considerations when integrating robot tutors into educational settings, with a focus on preserving the essential role of human educators and safeguarding the privacy and well-being of students.

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Title: "The Robot Brain Server: Design of a Human-Artificial Systems Partnership"
Author: Johan F. Hoorn

Summary:
In this paper, Johan F. Hoorn presents the concept of a "Robot Brain Server" and discusses its design as a means to establish a partnership between humans and artificial systems. The paper explores how this design can contribute to more effective and meaningful interactions between humans and robots.

Key Findings:
1. The Robot Brain Server:
   - The paper introduces the concept of the Robot Brain Server, which serves as the central processing and control unit for robots.
   - It emphasizes the importance of the Robot Brain Server in enabling more sophisticated and responsive interactions with robots.

2. Human-Artificial Systems Partnership:
   - The author explores the idea of creating a partnership between humans and artificial systems, with the Robot Brain Server acting as a crucial intermediary.
   - This partnership aims to enhance the collaboration and communication between humans and robots.

3. Design Principles:
   - The paper outlines the design principles for the Robot Brain Server, focusing on aspects such as adaptability, context awareness, and the ability to learn from interactions.

4. Enhanced Interactions:
   - It discusses how the Robot Brain Server can enable more natural, adaptive, and contextually relevant interactions between humans and robots, thereby improving the user experience.

5. Applications:
   - The paper touches upon potential applications of this design in various domains, including human-robot collaboration in healthcare, industry, and domestic settings.

Conclusions:
"The Robot Brain Server: Design of a Human-Artificial Systems Partnership" presents an innovative concept for improving the interactions between humans and robots through the design of a central processing unit, the Robot Brain Server. This design aims to create a more effective and meaningful partnership between humans and artificial systems, with applications in multiple domains. The paper highlights the potential of this concept to advance the field of human-robot interaction and emphasizes the need for adaptable, context-aware, and learning-capable systems to enhance user experiences and collaboration.

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Robot interaction
Paper Hypersummerizations

VU Amsterdam, year 23/24

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1. Robotics and artificial intelligence, Omankwu

robots more design, creation, operation, physical while AI more software driven to simulate human-like cognitive functions.

Both should be programmable

one can enhance the other

2. Design lessons from AI's two grand goals, schneiderman

Ai has as goal to mimic human cognitive processes

deep understanding of cognitive science and psychology is needed

Goal of useful applications of ai is to be practical and improve human lives

Human emulation doesn't always lead to the most practical solution

3. Social robots in therapy and care, Cifuentes

social robots increasing in amount in psychology health care

Non judgmental interaction

Also emerging as caregiver assistance for the elderly

Social robots have the ability to be more engaging and be more personalized

ethical concerns; Privacy, data security, and the potential for social robots to replace jobs

4. the in-between machine, Hoorn

robots are unique in that they are neither tools nor humans, they connect humans and the technological world

Critique that traditional models focus to much on hardware and functionality, undermining the potential of robots in society

robots value lies in their ability to collaborate, assist and augment human capabilities.

Human robot interaction optimization should have higher importance, thus an understanding between how humans and robots interact is crucial.

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5. Social Robots for Education, Belpaeme

Social robots are employed in educational contexts.

potential to enhance student engagement, with the ability to adapt to individual learning needs.

Physical embodiment in itself shows improved interactions compared to a digital companion.

Limitations: Cost, technical limitations, Teacher training to work with the bot

Again privacy concerns etc.

6. Alone together, Sherry Turkle

Technology has reshaped human connections

Illusion of connection, but ultimately isolated and disconnected

thus a need for face to face human interaction

7. How to build robots that make friends and influence people, Breazeal

Social intelligence; perceiving and responding to human emotions etc.

Importance of HRI and for the robots to engage in natural socially acceptable manner enabling the building of trust

Socially intelligent robots show a positive influence on human behavior

This social intelligence might influence humans decisions and emotions

8. Relationship development with humanoid social robots, Gambino - skipped

9. The influence of empathy in human robot relations, Leite

two forms of empathy; cognitive and emotional

cognitive empathy: understanding and recognizing human emotion

emotional empathy: capacity to express and respond to emotions

The presence of such empathy increases the quality of HRI also leading to more natural and supportive interactions

important things to include would be; facial expressions, tone of voice, etc.

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10. Coming of age : … neuroscience of semantics, meteyard - skipped

11. Modelling context awareness for a Situated Semantic Agent, Vossen

Basically NLP is good

Context awareness with relevant information from the environment being important (I.E. spatial, temporal, etc.)

Using NLP to respond appropriately

12. Robot Affect, Johan Hoorn

robot affect: emotional states and expressions of robots

Emotional capability important for more natural HRI

Analogy between amygdala and bloch sphere.

bloch sphere could be used to emulate the amygdala and manipulate a robots emotional state

this frame work could possible extend the scope of emotions robots can work with

This would in turn need complicated emotional recognition and expression

13. Reasoning about reasoning, stuhlmuller

Theory of mind: attributing beliefs, desires, and intentions to others allowing the individual to understand and predict the behavior of the other

Possible application in robots with predictive models to be more human like

14. Features of similarity, Tversky

how humans perceive and judge the likeness between different items can differ through the use of certain features such as dimensionality, complexity, directionality, and metric

15. Sustainable software, Sedano - skipped, basically just have good work flow an take overs

16. Robot tutors, smakman- skipped, feels like a repetition of a previous paper

17. Robot brain server, Hoorn - skipped, basically a control room that acts as the robots brain

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Robot interaction
Mock Test QnA

VU Amsterdam, year 23/24

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1. **Question 1:**
   Why do we need theories? (discussed in Lecture 1). Which of the following is NOT correct?
   A. Theories link concepts in order to explain something
   B. Theories present effective models of the complex interactions between humans and robots
   C. Theories reduce the complexity of reality
   D. Theories can be modeled
   Answer: B. Theories present effective models of the complex interactions between humans and robots.

2. **Question 2:**
   Lecture 1 (Introduction) discussed various complexities with key concepts in the field of Robot interaction. One of them is anthropomorphism. Which complexity has been discussed in view of robot design and research?
   A. Anthropomorphism is often understood as one's 'second self'.
   B. Anthropomorphism refers to individual perception but is mostly conceived as a design feature.
   C. Robots cannot have anthropomorphism, even not if they are humanlike.
   D. Anthropomorphism refers to a design feature but is mostly included in research as a processing variable.
   Answer: B. Anthropomorphism refers to individual perception but is mostly conceived as a design feature

3. **Question 3:**
   In Lecture 2 about similarities and differences between AI and robotics, the following paper was discussed: Omankwu, O. C., Nwagu, C. K., & Inyiama, H. (2017). Robotics and artificial intelligence: Differences and similarities. According to Omankwu et al., the similarity between AI and robots is that both AI and robots have to ...
   A. Interact with the physical world (e.g., through sensors and actuators)
   B. Be autonomous agents
   C. Be programmable
   D. Tackle learning, perception, problem-solving, language-understanding and/or logical reasoning – without sensors or actuators
   Answer: C. Be programmable.

4. **Question 4:**
   Lecture 3 discussed various applications of robots in healthcare and reviewed related research. One of the papers in your reading list (Hoorn et al., 2015) discussed 'modeling the wrong things' when modeling social robots. Why did they argue that? Which of the following answers is NOT correct?
   A. They found that imperfections in robot's design were okay.
   B. They found that no perfect humanlike resemblance was needed.
   C. They found it is important to focus attention to the users' unmet needs.
   D. They found that it helps to make copies of humans to make social robots work.
   Answer: D. They found that it helps to make copies of humans to make social robots work.

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5. **Question 5:**
Lecture 3 discussed various applications of robots in healthcare and reviewed related research. If you want to draw solid conclusions about the effects of a specific robot, which type of study do you need to do?
A. Focus groups with target users.
B. Co-creation with stakeholders
C. Observational case study.
D. Group comparison study.
Answer: D. Group comparison study.

6. **Question 6:**
What is the primary role of companion robots in healthcare (Lecture 3), particularly in their interaction with older adults?
A). To perform medical procedures and assist healthcare professionals in hospitals.
B). To monitor the health and safety of patients, especially those with special needs.
C). To initiate conversations, reduce perceived loneliness, and serve as conversational agents.
D). To provide educational play and entertainment.
Answer: C). To initiate conversations, reduce perceived loneliness, and serve as conversational agents.

7. **Question 7:**
One of the focus areas of the review of Belpaeme et al. (2018; Lecture 4: Robots in education) is the robot appearance and robot behavior. Which of the following statements related to this issue is true?
A. Almost any strategy or social behavior of the robot aimed at increasing learning outcomes has a positive effect (both cognitive and affective).
B. It is unlikely that just the physical presence of the robot promotes social behaviors in the learner that, in turn, foster learning and create a positive learning experience.
C. Straightforward forms of personalization, such as using a child's name or referencing personal details within an educational setting, do not enhance user perception of the interaction.
D. Because of the complexity of the experimental protocols of the studies reviewed by Belpaeme et al. (2018), they were not able to make a direct comparison between different robots used for learning.
**Answer: A. Almost any strategy or social behavior of the robot aimed at increasing learning outcomes has a positive effect (both cognitive and affective). **

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8. **Question 8:**

Lecture 4 (Robots in education) discussed a review and meta-analysis of selected papers that empirically tested the effectiveness of robot tutors. Results were promising, yet, many studies suffer from methodological limitations. One of them is that many studies are heavily underpowered. What does it mean?

A. The effect size is too small for reliable statistical conclusions.

B. The control group is lacking for reliable statistical tests.

C. The sample size is too small for reliable statistical tests.

D. The data are too heterogeneous for reliable statistical conclusions.

Answer: C. The sample size is too small for reliable statistical tests.

9. **Question 9:**

Lecture 5 discussed various psychological perspectives on human-robot relationships. Sherry Turkle (2011) discussed machine-mediated relationships as an illusion of companionship. What can be seen as problematic in this paper of Turkle?

A. Problematic in Turkle is the perspective of robots as substitutes for humans, for example, in referring to how mediated connections have evolved over the past decade.

B. Problematic in Turkle is the psycho-analytic perspective wherein robots are perceived as illusions for companionship because people are in need of deeper, meaningful connections.

C. Problematic in Turkle is the perspective of robots as illusions for companionship because today, people are used to mediated connections.

D. Problematic in Turkle is the perspective of robots as substitutes for humans because people know how to compensate for mediated connections.

**Answer:

B. Problematic in Turkle is the psycho-analytic perspective wherein robots are perceived as illusions for companionship because people are in need of deeper, meaningful connections.**

10. **Question 10:**

From lecture 6 (Daniel Preciado; paper Damholt et al.'s scoping review), what does the term "anthropomorphism" primarily involve?

A) The study of human origins and evolutionary history.

B) The design of robots with human-like physical characteristics and behaviors.

C) The ethical considerations of integrating robots into society.

D) Attributing human-like qualities or characteristics to non-human entities, including robots.

Answer: D) Attributing human-like qualities or characteristics to non-human entities, including robots.

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11. **Question 11:**
What is the difference between a Two Stream and One Stream fusion transformer model to combine text and images?
A. One-stream models can use pretrained models and Two-stream models cannot.
B. One-stream uses separate pretrained models for text and images and Two-stream models require pretraining using combinations of text and images.
C. Two-stream models can use separate pretrained models for text and images and One-stream models require pretraining using combinations of text and images.
D. Two-stream models can use pretrained models and One-stream models cannot.
Answer: C. Two-stream models can use separate pretrained models for text and images, and One-stream models require pretraining using combinations of text and images.

12. **Question 12:**
In lecture 6, for the empathy implementation in the iCat scenario described in Leite et al.'s paper (The influence of empathy in human–robot relations, 2013), the authors took inspiration from human-human interactions in educational settings. Which characteristics of empathetic teachers were implemented in the iCat?
A. Verbal responses, voice intonation.
B. Context-relevant facial expressions, verbal responses.
C. Body language, voice intonation.
D. Context-relevant facial expressions, body language.
Answer: B. Context-relevant facial expressions, verbal responses.

13. **Question 13:**
As discussed in lecture 7 (NLP), Contextual Word Embeddings or Large Language Models can disambiguate the meaning of a word such as "star" in context. Which of the following is correct?
A. By creating separate word embeddings for each meaning of "star".
B. By memorizing the meanings of the word "star" as different embeddings.
C. The word "star" will be represented the same regardless of context.
D. By combining the word embedding of "star" with the embeddings of words surrounding it.
Answer: D. By combining the word embedding of "star" with the embeddings of words surrounding it.

14. **Question 14:**
From lecture 7 (Piek Vossen): What is the difference between a Two Stream and One Stream fusion transformer model to combine text and images?
A. One-stream models can use pretrained models and Two-stream models cannot.
B. One-stream uses separate pretrained models for text and images and Two-stream models require pretraining using combinations of text and images.
C. Two-stream models can use separate pretrained models for text and images and One-stream models require pretraining using combinations of text and images.
D. Two-stream models can use pretrained models and One-stream models cannot.
Answer: C. Two-stream models can use separate pretrained models for text and images, and One-stream models require pretraining using combinations of text and images.

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15. **Question 15:**

Referring to lecture 8 on language understanding and generation: The Leolani system (Vossen et al. 2019) stores everything you tell her and uses a theory of mind model. Which of the following is correct for the Leolani model?

A. The theory of mind model enables her to store conflicting information

B. The theory of mind model enables her to decide what is true

C. The theory of mind model orders facts according to their probability

D. The theory of mind model explains that people think the robot has a mind of its own

Answer: C. The theory of mind model orders facts according to their probability.

16. **Question 16:**

In Lecture 9 about modeling for theory development, we discussed the k-Nearest-Neighbor algorithm to model similarity estimates with Euclidian distance. Using the k-NN algorithm to model how people perceive similarity between them faces the problem that each instance is valued the same although similarity estimates are based on matching feature sets and can be asymmetric (Tversky, 1977). A smart approach could be to assign weights to the contributions of the neighbors, so that the nearer neighbor adds more to the average than distant ones (e.g., 1 divided by the distance to the neighbor). What is the problem with this approach (most complete and precise answer counts)?

A. This is not how psychology works.

B. The k-NN algorithm is sensitive to the local structure of the data.

C. The k-NN algorithm relies on distance, so normalizing the training set should be done for higher accuracy of the similarity estimate.

D. The k-NN algorithm represents feature matching in Euclidian space. The weights are spatial, not cognitive or affective.

Answer: D. The k-NN algorithm represents feature matching in Euclidian space. The weights are spatial, not cognitive or affective.

17. **Question 17:**

For robot interaction, an AI developer has to choose from a number of approaches to fuzzy intersection. She can choose from: 1 – Min{1,((1 - A)P + (1 - B)P)1/P} (Yager) and Min{1,[b• (1 - A]P / [a• (1 - B]P} (Baldwin et al.) In Lecture 10 about modeling for application and practice, we discussed what criterion to use to make such choice:

A. Engineering convenience

B. Mathematical simplicity

C. User satisfaction

D. Fidelity to psychological theory

Answer: D. Fidelity to psychological theory. (Not sure if correct)

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18. **Question 18:**

De-humanizing is one of the issues discussed by Serholt et.al. (2017; lecture 11 – Ethics of robots in the classroom). Which of the statements below does NOT reflect the perceptions of teachers in Serholt's study related to De-humanizing.

A. Robots cannot interact on the same emotional plane as humans

B. Robots were viewed in social terms as an additional teaching tool to guide the teacher as to how to create the optimal learning environment.

C. People's use of language is highly affected by technology, children might mimic robots

D. If children were to engage in destructive physical actions, these could ultimately legitimize and encourage violence in human interactions.

Answer: B. Robots were viewed in social terms as an additional teaching tool to guide the teacher as to how to create the optimal learning environment.

19. **Question 19:**

One of the values discussed in Smakman and Konijn (2020; lecture 11 – Ethics of robots in the classroom) is Efficiency. Related to this value we discussed three issues during our lecture. Which one of the issues below did NOT relate to the value of Efficiency?

A. Voice and speech recognition

B. The response to social context is still lacking

C. Negative impact on the concept of friendship

D. The cost of the robot

Answer: C. Negative impact on the concept of friendship.

In lecture 12 about ethics, privacy, and moral concerns with respect to robots, a moral reasoning system was presented as designed by: Pontier, M. A., Widdershoven, G., Hoorn, J. F., Van Gelder, J. L., & De Vries, R. (2014). Moral Coppélia: Affective moral reasoning with twofold autonomy and a touch of personality. Machine Ethics in the Context of Medical and Care Agents (MEMCA-2014), 1-4 April 2014. AISB-50, Goldsmiths, University of London.

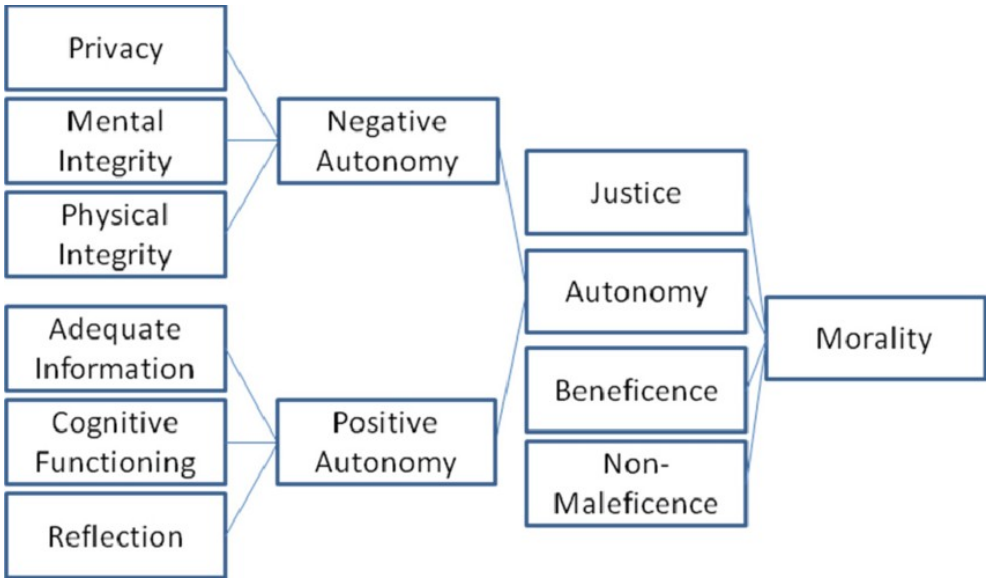
In that paper, positive autonomy of the user was calculated by the robot as:

A Information * Cognitive_Functioning * Affect

B Information * Mental_Integrity * Reflection

C Privacy * Cognitive_Functioning * Reflection

D Information * Cognitive_Functioning * Reflection



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1. **What is the main argument in the paper of Turkle (2011)?**
- In Turkle's paper, the main argument is that machine-mediated relationships, such as interactions with robots and technology, can create an illusion of companionship. Turkle suggests that as people increasingly turn to technology for social interactions, they might be substituting meaningful human relationships with more shallow or illusory connections through technology.

2. **What is meant with ‘modeling the wrong things’? Report at least two arguments.**
- 'Modeling the wrong things,' as discussed in Hoorn et al. (2015), refers to the practice of focusing on aspects of social robots that may not be essential for achieving their intended goals. Two arguments from the paper are:
  - Imperfections in a robot's design or a lack of perfect human-like resemblance are acceptable as long as the robot effectively fulfills its intended role. Thus, striving for an excessively human-like appearance is unnecessary.
  - It is crucial to direct attention to users' unmet needs and requirements rather than attempting to create robotic replicas of humans.

3. **What are several understandings or uses of the concept of anthropomorphism, following the scoping review of Damholdt?**
- Damholdt's scoping review of anthropomorphism highlights various understandings and uses of the concept, including:
  - Anthropomorphism as a design feature, where robots are intentionally designed to resemble or act like humans.
  - Anthropomorphism as a processing variable in research, where it is introduced to investigate its impact on human-robot interaction and user experience.
  - Anthropomorphism as a psychological perspective, where individuals attribute human-like qualities and intentions to robots.

4. **What was the main aim in the study of Breazeal & Scassellati (1999) in developing robot Kismet?**
- The main aim of Breazeal and Scassellati in developing robot Kismet was to test the Theory of Mind (ToM). They wanted to create a robot (Kismet) that could interact socially with a human caregiver, and the human caregiver should believe that Kismet has intentionality, beliefs, and desires. The goal was to explore whether the robot could establish social interactions with humans by implementing pro-social responses.

5. **What is ‘para-social interaction’? (def./key aspects + example), and how is it similar/different from HRI?**
- Para-social interaction refers to the psychological phenomenon in which individuals form one-sided, emotional connections with media personalities or fictional characters. This interaction is similar to human-robot interaction (HRI) in that it involves individuals forming emotional connections with non-human entities. However, para-social interaction is typically directed toward media figures, while HRI involves interaction with physical or virtual robots. In para-social interaction, individuals often feel a personal connection with a media figure or character, even though there is no real reciprocity or two-way interaction, while in HRI, the interaction involves real-time communication with a robot.

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6. **What are the three interrelated aspects of language grounding (Vossen et al. 2019) in physical real-world contexts?**

- Vossen et al. (2019) discuss three interrelated aspects of language grounding in physical real-world contexts:
  - **Contextual Embedding:** This involves the integration of contextual information, such as spatial, temporal, and situational cues, to understand and generate language.
  - **Episodic Memory:** It refers to the storage and retrieval of specific past experiences and events as they relate to language understanding and generation.
  - **Theory of Mind (ToM):** ToM enables robots to understand the beliefs, intentions, and perspectives of humans, allowing for more effective and contextually appropriate communication in physical real-world environments.

These aspects are crucial for enabling robots to engage in meaningful and contextually relevant language interactions with humans in the physical world.

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